Exercise 10

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AIM: The aim is to create data visualizations, such as line charts and doughnut charts, for a student performance analysis system using JavaScript and Chart.js.

PROCEDURE:

Step 1: Set Up Your HTML File Create an HTML file to define the structure, including canvas elements where the charts will be rendered, and link the Chart.js library.

HTML (index2.html**):**

HTML

```html
<!DOCTYPE html>

<html>

<head>

  <title>Student Performance Charts</title>

  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

  <style>

    body { font-family: Arial, sans-serif; background-color: #f4f4f4; text-align: center; }

    h2 { color: #222; }

    canvas {

      display: block;

      margin: 20px auto;

      background-color: #fff;

      border: 1px solid #ddd;

      border-radius: 6px;

      box-shadow: 0 2px 6px rgba(0, 0, 0, 0.1);

    }
```

```
  </style>

</head>

<body>

  <h2>Student Performance Analysis</h2>


  <h3>Marks Progress (Line Chart)</h3>

  <canvas id="lineChart" width="400" height="200"></canvas>


  <h3>Subject-wise Performance (Doughnut Chart)</h3>

  <canvas id="doughnutChart" width="400" height="200"></canvas>


  <script src="script1.js"></script>

</body>

</html>
```

Step 2: Create the JavaScript File for Charts Create a JavaScript file to define the datasets for test marks and subject distribution, and initialize the charts using the Chart.js constructor.

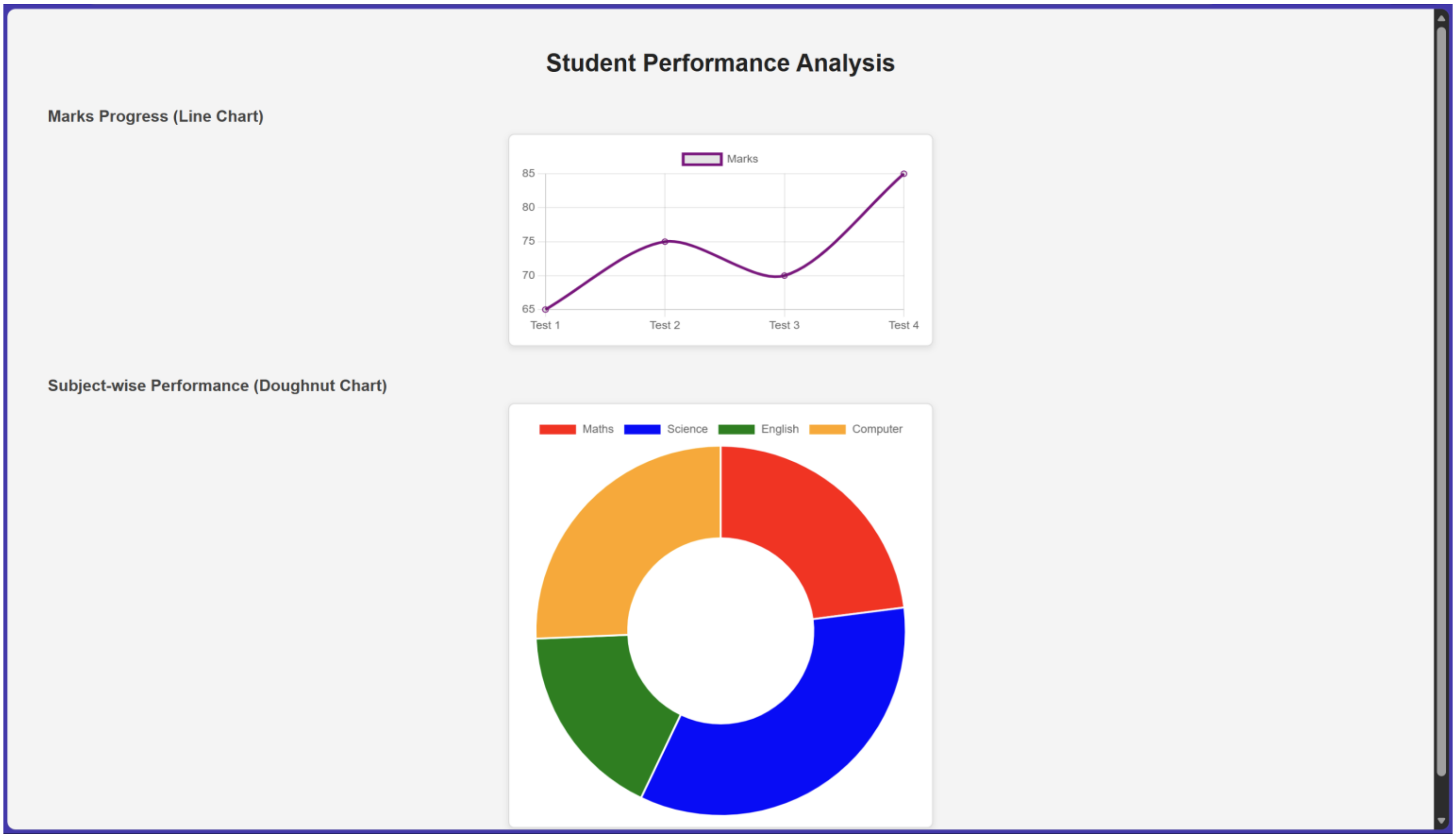
JavaScript (script1.js**):**

JavaScript

```
// Line Chart - Marks Progress

const lineCtx = document.getElementById('lineChart').getContext('2d');

const lineChart = new Chart(lineCtx, {

  type: 'line',

  data: {

    labels: ['Test 1', 'Test 2', 'Test 3', 'Test 4'],

    datasets: [{

      label: 'Marks',

      data: [65, 75, 70, 85],

      borderColor: 'blue',

      fill: false,
```

```
      tension: 0.3
    }]
  },
  options: { responsive: true }
});

// Doughnut Chart - Subject Performance
const doughnutCtx = document.getElementById('doughnutChart').getContext('2d');
const doughnutChart = new Chart(doughnutCtx, {
  type: 'doughnut',
  data: {
    labels: ['Maths', 'Science', 'English', 'Computer'],
    datasets: [{
      label: 'Marks Distribution',
      data: [85, 75, 80, 90],
      backgroundColor: ['red', 'blue', 'green', 'orange']
    }]
  },
  options: { responsive: true }
});
```

OUTPUT:



RESULT: The data visualizations for student performance were successfully created using JavaScript and Chart.js, providing a clear graphical representation of marks progress and subject-wise distribution.